

Retail Business Performance Analysis

End-to-end project: Python data cleaning, MySQL analysis and Power BI dashboards

1. Introduction

Retail businesses generate large volumes of transaction data, but raw records rarely answer practical questions: which product categories really earn money, which regions and seasons drive growth, and how long stock sits before it sells. This project analyses 12,000 retail orders placed between January 2024 and December 2025 across 19 Indian states, 5 regions, 5 product categories (19 sub-categories, 285 products) and 2,485 customers, sold through Online, Marketplace and Store channels. The goal was to turn the raw dataset into clear, decision-ready insight on sales, cost, profit and inventory.

2. Abstract

The project follows a complete analytics workflow. The raw CSV was cleaned and enriched in Python (Pandas), loaded into MySQL where eleven analytical queries measured business, category, regional, seasonal, monthly and inventory performance, and finally visualised in an interactive three-page Power BI dashboard. The business generated **5.17M in sales** against **4.16M in cost**, leaving **1.01M in profit** (an overall margin of 19.6% on total sales; the dashboard card shows 23.22, the average of order-level margins) from 36,153 units sold. Technology is the largest profit contributor (0.40M) but has the thinnest margin (14.9%), while Personal Care has the highest margin (31.3%) on the smallest profit. South and West lead regional profit, and seasonal performance is very even, with Summer slightly ahead.

Total Sales	Total Cost	Total Profit	Margin (Profit/Sales)	Units Sold	Avg Inventory Days
5.17M	4.16M	1.01M	19.6%	36,153	39.3

3. Tools Used

Tool	Purpose
Python (Pandas, NumPy)	Loading, quality checks and feature engineering of the dataset
Matplotlib / Seaborn	Exploratory visual checks inside the notebook
Jupyter Notebook	Documented, reproducible cleaning workflow
Microsoft Excel (.xlsx)	Storing the cleaned dataset for loading and review
MySQL (SQL)	Aggregation and business analysis in the <i>retail_analysis</i> database
Power BI	Interactive dashboards with slicers, KPI cards, charts and tables

Dataset fields

The cleaned file has 24 columns: Order_ID, Order_Date, Customer_ID, Region, State, Category, Sub_Category, Product, Quantity, Unit_Price, Discount, Sales, Cost, Profit, Inventory_Days, Payment_Method, Sales_Channel and Season, plus six derived fields: Profit_Margin, Gross_Sales, Discount_Amount, Year, Month and Month_Name.

Sample key results by category

Category	Sales	Profit	Margin	Avg Inv. Days
Technology	2.66M	0.40M	14.9%	39.5
Furniture	1.35M	0.30M	22.3%	38.7
Home & Kitchen	0.62M	0.16M	26.2%	39.8
Office Supplies	0.32M	0.09M	26.9%	38.8
Personal Care	0.21M	0.07M	31.3%	40.1

4. Steps Involved in Building the Project

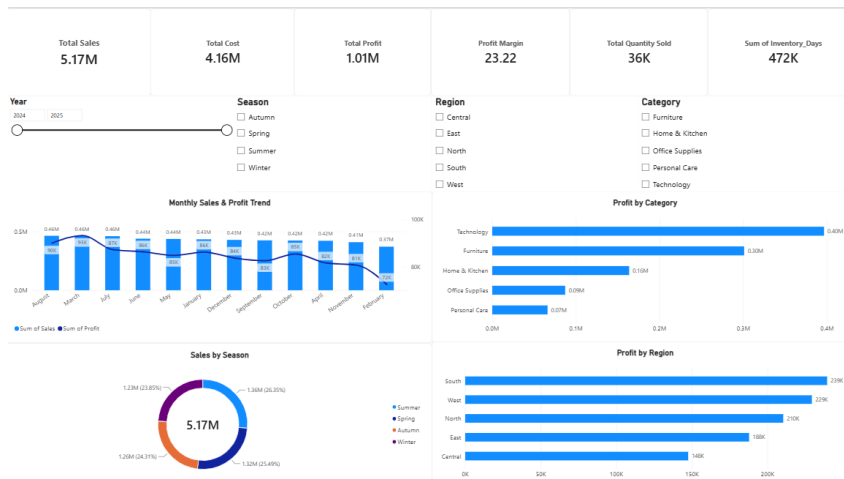
Step 1 - Data loading and inspection. Loaded the raw CSV in Jupyter and checked shape (12,000 rows), data types, missing values, duplicate rows and duplicate Order IDs.

Step 2 - Data cleaning. Converted Order_Date to datetime and reviewed the numeric columns (Quantity, Unit_Price, Discount, Sales, Cost, Profit, Inventory_Days) with summary statistics.

Step 3 - Feature engineering. Created Profit_Margin, Gross_Sales, Discount_Amount, Year, Month and Month_Name to support deeper analysis; saved the result as the cleaned dataset.

Step 4 - SQL analysis. Loaded the data into MySQL table *retail_data* and wrote 11 query blocks: overall KPIs, category, sub-category, region, state, season, category-by-season, monthly trend, inventory by category, slow-moving products and inventory days vs profit.

Step 5 - Dashboard design. Built a Power BI report with KPI cards and slicers (Year, Season, Region, Category, Sub-Category) across three pages: Executive Overview, Product Performance, and Seasonal and Inventory analysis.



Dashboard page 1 - Executive Overview

KPI cards, monthly sales and profit trend, profit by category and region, and sales share by season.

Key findings from the analysis

- Technology** earns the most profit (0.40M, 51% of sales) but its 14.9% margin is lowest; **Personal Care** has the best margin (31.3%) but the smallest profit (0.07M).
- Regions:** South (239K) and West (229K) lead profit; Central is lowest (148K). Margins are almost identical (about 19.5-19.7%) in every region, so differences come from sales volume.
- Seasons:** Summer is highest (1.36M sales, 0.26M profit) and Winter lowest (1.23M sales), a spread of only about 10%, so demand is fairly stable across the year.
- Sub-categories:** Storage (121K), Printers (108K), Accessories (100K) and Phones (98K) top profit; Grooming and Hair Care are lowest (16K each).
- Inventory:** average holding is about 39 days, with little difference between categories (38.7-40.1). Technology and Furniture products with higher inventory days tend to show higher profit, while Personal Care and Office Supplies stay low in profit regardless of inventory days.

5. Conclusion

The project shows how Python, SQL and Power BI combine into a reliable pipeline: clean and enrich the data, analyse it with repeatable queries, and communicate the results visually. The analysis indicates a healthy business with a 19.6% margin, but profit depends heavily on Technology, which sells in volume at low margins. Recommended actions are to protect Technology pricing and cost, grow high-margin Personal Care and Home & Kitchen lines, review low-profit items such as Grooming and Hair Care products, and monitor slow-moving products to free up working capital. The approach can be extended with customer segmentation, discount-impact analysis and sales forecasting.